Series 0 – *Reminders on Basics of Algorithms*

Exercise 1

To perform the cubic root of a real number $A \geq 0$ we can use the following iterative method : We take an initial value $x1=A/3$ then we search for successive items $x2$, $x3$,... of the sequence defined by:

$$x_{n+1} = \frac{2x_n + \frac{A}{x_n^2}}{3}$$

After each iteration, we compare x_{n+1} to x_n and we stop the process when $|x_{n+1} - x_n| < 10^{-6}$.
In this case, x_n represents the cubic root of A. Write an algorithm to provide the cubic root of a real number A using the described method below.

Exercise 2

Let T be a one-dimension array containing N integers (where N≤100). Write an algorithm to delete all the null values from the array T.

Exercise 3

Consider the following procedures, where Plist is a pointer to a node of a list of integers:

```
Procedure modifList1(I/ L :Plist) ;
Begin   While L≠Nil Do L^.Data ← 2*L^.Data ;  L ← L^.Next ; Endwhile ; End ;

Procedure modifList2(IO/ L :Plist) ;
Begin
   While L≠Nil Do L^.Data ← 2*L^.Data ;  L ← L^.Next ; Endwhile ;
End ;

Procedure modifList3(IO/ L :Plist) ;
Var V :Plist ;
Begin   V← L ; While V≠Nil Do V^.Data ← 2*V^.Data ;  V ← V^.Next ; Endwhile ; End ;

Procedure NextList1(I/ L :Plist) ;
Begin   If  L≠Nil  Then L ← L^.Next ; Endif ; End ;

Procedure NextList2(IO/ L :Plist) ;
Begin   If  L≠Nil  Then L ← L^.Next ; Endif ; End ;
```

Consider the list **L**= {1, 2, 3}. Provide the content of the list **L** after the execution of each of the procedures below.

Exercise 4

Let **L** be a list of words of maximal size 25 characters.

1- Provide the declaration of this list.
2- Write a function **Palindrome** to check if a given word is palindrome.
3- Write a procedure **DetachPal** that provides the address of the first palindrome word of the list **L** and removes it from the list without **deleting it**.
4- Using the procedure **DetachPal**, write a procedure **ListPal** to create a list **LP** containing all the palindrome words of the list **L** in the inverse order of their appearing in the list **L** (**without creating a new list**).